

PROGRAM ASSESSMENT REPORT

v2023.21 current

Tracking data

CIM code	BS-CHE
Program name	Chemical Engineering
academic year	2024-2025
Program assessment coordinator	Victor HugoPerez-Luna
Assessment plan year	2025
No assessment this year per plan	☐
LO(s) assessed	5, 6, and 7 ABET Learning Outcomes
Presented at program authority meeting date:	To be presented at faculty meeting on Nov. 10, 2025 11/10/2025
<i>Please attach minutes</i>	Minutes will be provided after Nov. 25, 2025

Methodology

Student outcomes 5 – 7 from ABET are being evaluated for the 2024- 2025 academic year. ABET establishes the following 7 student outcomes for engineering programs:

The Engineering Criteria Student Outcomes (1)-(7) are:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. an ability to communicate effectively with a range of audiences.
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

The Department of Chemical and Biological Engineering has a set of 9 Student Outcomes linked to ABET Engineering Criteria Student Outcomes 1 – 7. The Student Outcomes of the ChBE Department are to develop individuals who:

- I.** possess fundamental knowledge of mathematics, the basic sciences and engineering (*ABET Engineering Criteria Student Outcome 1*)
- II.** possess fundamental knowledge of chemical and biological engineering (*1*)
- III.** possess the necessary skills to design processes that are safe, economic, technically sound and environmentally benign (*2*)
- IV.** have been trained to fully utilize recent advances in computational technologies in their work (*1, 2, and 6*)
- V.** possess a good familiarity with experimental and analytical methods and techniques (*1, 2, and 6*)
- VI.** have been trained to fully utilize modern communication technologies and who communicate effectively (*3*)
- VII.** possess leadership skills and the ability to work in intra-disciplinary and multi-disciplinary teams (*5*)
- VIII.** have an understanding and respect for professional ethics and possess an awareness of the present-day societal issues of relevance to chemical engineers (*4*)
- IX.** have the ability and desire to further their education through the process of “Life Long Learning” and possess the necessary skills to be involved in research and development activities, if they so desire (*7*)

The mapping of ChBE Student Outcomes **I – IX** to ABET Engineering Criteria Student outcomes 1 – 7 are given in the following table:

Student Outcome	1	2	3	4	5	6	7
I (Math and Basic Science)	X						
V (Experimental/Analytic Skills)	X	X				X	
III (Design of Processes)		X					
VII (Leadership / Teamwork)					X		
II (Fundamentals of CHE)	X						
VIII (Ethics / Societal Issues)				X			
VI (Communication Skills)			X				
IX (Life Long Learning)							X
IV (Computational Skills)	X	X				X	

Each core course of the Chemical Engineering Program has a specific set of Student Learning Objectives (SLOs) and each individual SLO within each course is related to one or more of the **I-IX** Student Outcomes (and 1 – 7 ABET Engineering Criteria Student Outcomes). The ChBE department keeps this information with the syllabus of each core CHE course. The syllabi for courses evaluated in this cycle are presented later in this report.

Each semester, and for each course, copies of exams and solutions, projects, and or homeworks used in the course are collected. Handouts and notes given in each course are also collected and archived with each course’s evaluation. Samples typical of the higher, lower and average student scores for quizzes,

papers, homework, exams and midterms are collected and kept in a folder for each course evaluated. During the last week of the semester students are given a student assessment form to evaluate the level at which each SLO was met in the course and their individual mastery level for each SLO.

The following are the forms used to assess each course. These constitute the standard set of procedures by which the ChBE department assesses its educational goals.

[illegible]

Refer to the course syllabus handed out at the beginning of the semester and provide a rating for each objective in each column of the following table: (1-low, 5-High)

[illegible]

Please comment on the ratings given above:

[illegible]

Illinois Institute of Technology
Department of Chemical and Biological Engineering
Students Course Assessment Tally Sheet – Form 1

Course: **ChE** ____ Semester: **_ Fall** **Year 20__** Number of students _____
_ Spring

G-1. Tally the # of students in each category for each of the course SLOs
(from Student Course Assessment Forms)

(a) Level at which the SLO has been met in this course?

	1	2	3	4	5	$\geq 3, (\%)^*$
SLO 1						
SLO 2						
SLO 3						
SLO 4						
SLO 5						
SLO 6						
SLO 7						
SLO 8						
SLO 9						
SLO 10						
SLO 11						
SLO 12						

*Also record in Form-3 Column 1

(b) Students' level of mastery after taking this course?

	1	2	3	4	5	$\geq 3, (\%)^{**}$
SLO 1						
SLO 2						
SLO 3						
SLO 4						
SLO 5						
SLO 6						
SLO 7						
SLO 8						
SLO 9						
SLO 10						
SLO 11						
SLO 12						

**Also record in Form-3 Column 2

Instructor's Signature _____ Date _____

Illinois Institute of Technology

Department of Chemical and Biological Engineering

FACULTY COURSE SELF-ASSESSMENT – Form 2

Course: **ChE** ____ Semester: **_ Fall** **Year 20--** Number of students____
_ Spring

- How many students earned each grade? A ____ B ____ C ____ D ____ E ____
- Indicate the # of students in each category for each of the course SLOs.

1=Unacceptable, 2= Need Improvement, 3= Adequate, 4=Met Expectation, 5= Achieved Mastery

	Assessment Method (HW, test, ...)	1	2	3	4	5	≥3, (%)***
SLO 1							
SLO 2							
SLO 3							
SLO 4							
SLO 5							
SLO 6							
SLO 7							
SLO 8							
SLO 9							
SLO 10							
SLO 11							
SLO 12							

*** Also record in Form-3 Column 3

1- In your opinion, were >70% the students adequately prepared in mathematics?
 (Yes/No).____

If "No", identify the subject and indicate the # of students in each category for each subject.

1=Unacceptable, 2= Need Improvement, 3= Adequate, 4=Met Expectation, 5= Achieved Mastery

Math Subject	1	2	3	4	5

2- In your opinion, were >70% the students adequately prepared in the basic sciences? (Yes/No). ____

If "No", identify the subject and indicate the # of students in each category for each subject.

1=Unacceptable, 2= Need Improvement, 3= Adequate, 4=Met Expectation, 5= Achieved Mastery

Basic Science Subject	1	2	3	4	5

- In your opinion, were >70% the students adequately prepared in the fundamentals of Chemical Engineering? (Yes/No). _____

If "No", identify the subject and indicate the # of students in each category for each subject.

1=Unacceptable, 2= Need Improvement, 3= Adequate, 4=Met Expectation, 5= Achieved Mastery

CHE Course or Subject	1	2	3	4	5

- Describe all activities in this course that are conducive of the "Life Long Learning" process. How do you rate the student's accomplishments in this regard? (Use additional pages if needed)

- If you were to teach the course again, what changes would you make? (Use additional pages if needed)

- Other Comments

Instructor's Signature: _____

Date: _____

Course: _____ Semester: **_ Fall Year ____**
 _ Spring

- | | Students' Assessment
(Objective Met)* | Students' Assessment
(Mastery Level)** | Instructor's Assessment*** |
|--------|--|---|----------------------------|
| SLO 1 | | | |
| SLO 2 | | | |
| SLO 3 | | | |
| SLO 4 | | | |
| SLO 5 | | | |
| SLO 6 | | | |
| SLO 7 | | | |
| SLO 8 | | | |
| SLO 9 | | | |
| SLO 10 | | | |
| SLO 11 | | | |
| SLO 12 | | | |

Instructor's Signature _____ **Date** _____

Illinois Institute of Technology
Department of Chemical and Environmental Engineering
Graduating Senior Exit Interview form

Interviewer: Dept. Chair

Year:

Semester:

Student Name:

- **Background Information** (student name., degree completed, GPA, male/female, ethnicity[optional], specialization or minor[if any], ...)
- **Student Assessment of General Education Courses**(Math, Sciences, Humanities and soc. sci.)
- **Student Assessment of Technical ChE Courses** (what are dept. strengths and weaknesses?)
- **Communication, Teamwork, Life-long learning, leadership** (does student feel adequately prepared in these areas?)
- **Utilization of Computers in ChE Education** (does student feel adequately prepared in this area?)
- **Integration of Material among ChE Courses** (is it evident or absent? Give specific examples)
- **Breadth of education and awareness of present-day issues of importance to ChEs**(describe student experience in this regard, especially considering specialization and/or minor)
- **Preparation for career in industrial R&D or academic research** (student preference, attending grad. school or not?, examples of research-based activities)
- **Feedback on student outcomes.** Please provide your feedback on attainment of each student outcome I-IX.
- **Suggestions for improvement in the curriculum**

Coop Student Self-Assessment Form

Supervisor Name _____

First

Company Location_____

State

Would you please check the appropriate boxes for the following questions? If no basis for judgment, check NA.

1. How well did IIT education prepare you for the experiment/analytical skills required in your Coop Experience?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ NA ☐

2. How well was your mastery of communication including oral & written presentations at Coop?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ NA ☐

3. How did your leadership/teamwork experience at IIT measure up to the expectations at Coop?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ NA ☐

Comments:

Department of Chemical Engineering Illinois Institute of Technology

Coop Supervisor Assessment Form

Coop Student Name _____

Supervisor Name _____

Last

First

Company Name _____

Company Location _____

City

State

Dear Supervisor,

Would you please check the appropriate boxes for the following questions? If no basis for judgment, check NA.

Please answer based on a scale of 1-5 (1-lowest, 5-highest):

1. Student mastery level of experiment/analytical skills

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ NA ☐

2. Students skills of communication including oral & written presentations

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ NA ☐

3. Student leadership/teamwork qualities

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ NA ☐

Comments:

Department of Chemical and Biological Engineering
Evaluation of Undergraduate Research Experience

Every student participating in an undergraduate research project under a ChBE faculty member is required to complete the following form. The scale ranges from 1-5, with 1 meaning strongly disagree with statement, and 5 meaning strongly agree with statement.

For the Student:

1. The undergraduate research project has improved my knowledge of experimental and/or analytical methods and techniques

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

2. The undergraduate research project has trained me to utilize modern communication technologies and improved the efficacy of my communication

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

3. The undergraduate research project has instilled in me the ability and desire for “life-long learning” and given me the skills necessary for continued involvement in research and development

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

For the Supervisor:

1. The undergraduate research project has improved the student’s knowledge of experimental and/or analytical methods and techniques

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

2. The undergraduate research project has trained the student to utilize modern communication technologies and improved the efficacy of his/her communication

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

3. The undergraduate research project has instilled in the student the ability and desire for “life-long learning” and given him/her the skills necessary for continued involvement in research and development

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Supervisor Name:

Course No:

Semester and Year:

Research Project Topic:

AIChE Faculty Advisor Assessment Form

Academic year: 20__/20__ Faculty Advisor Name: _____

- 1) Please estimate how many students were involved in the Chapter _____
- 2) Please indicate how many students were involved in the Executive Board _____
- 3) Please rate the activity level of the Chapter as a whole:

not active poor acceptable highly active exceptional

- 4) Please rate the Teamwork of the Executive Board by indicating the number of officers with the following performance.

	poor	acceptable	high
Number of officers in each category ---->	_____	_____	_____

- 5) Please rate the Leadership of the Executive Board by indicating the number of officers with the following performance.

	poor	acceptable	high
Number of officers in each category ---->	_____	_____	_____

- 6) Please estimate how many events / speakers were sponsored by the Chapter _____
- 7) Please estimate the average number of ChE students attending each events _____
- 8) How many of these events / speakers discussed Ethical Issues _____
- 9) How many of these events / speakers discussed Societal Issues _____

Please attach any supporting documents you think would be useful. Including:

- The annual report sent to AIChE national.
- Any Newsletters generated by the Chapter.
- Copies of minutes from officer or general meetings.

AIChE Student Conference Debriefing Form

Student Name: _____

Conference Name: _____ Location: _____

Conference Date: _____

1) Were there any workshops offered at the conference? (Y / N)

If so, how many, _____ and please summarized the topics presented

2) How many workshops did you attend? _____

3) Of the workshops you attended, how many discussed Ethical Issues? _____

4) Of the workshops you attended, how many discussed Societal Issues? _____

5) Please highlight the specific Ethical or Societal Issues discussed in these workshops:

6) Please highlight any other noteworthy events or activities you participated in at the conference. (Use the back page or attach additional pages if necessary.)

In this assessment cycle, the following courses were evaluated according to the rubrics established within the self study report of the ChBE department for ABET.

CHE 202 Material and Energy Balances

CHE 302 Heat and Mass Transfer

CHE 311 Foundations of Biological Sciences for Engineers

CHE 317 Chemical and Biological Engineering Lab I

CHE 351 Thermodynamics I

CHE 418 Chemical and Biological Engineering Lab II

CHE 451 Thermodynamics II

CHE 494 Process Design I

The metrics normally employed in the creation of the self study report for ABET to determine whether the ABET Engineering Criteria Student Outcomes (5-7) and ChBE Student Outcomes **IV, V, VII** and **IX**, are being met are presented in the following pages. In this interim report the results of exit interviews, alumni surveys, surveys about experiences for co-op and summer internships, IPRO assessments, undergraduate research, and evaluation of the AIChE student chapter are not being included. The methodology for placing students in the 3 assessment categories {**D** *Does not meet expectations*, **M** *Meets expectations* **M**, **P** *Proficient*} is given only for a subset of courses and the mechanisms for making this assessment are described in each evaluation. This is an area where data collection and assessment in the ChBE department has not been standardized and the process for doing so will be implemented for future evaluation cycles. Artifacts of student's work are also presented as a small subset. These are kept in file in the ChBE department and can be accessed for examination upon request.

Student Outcome IV

Outcome IV: Developing individuals who have been trained to fully utilize recent advances in computational technologies in their work.

- Metrics:** The following were used as targets of performance for outcome IV.
- 1- The instructors' assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
 - 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
 - 3- 75% of the students respond favorably to question 5 on the exit interview form.
 - 4- 80% of alumni surveyed expressed satisfaction with this outcome.

Supporting Courses and Activities:

Courses/Activity	Elements Contributing to Outcome
CHE 100	SLO 1-6
CHE 101	SLO 2,4,5
CHE 239	SLO 1-10
CHE 302	SLO 6
CHE 433	SLOs 1-8
CHE 423	All SLOs
CHE 435	SLOs 10-12
CHE 494	SLOs 3 and 7
CHE 406	SLO 6
CHE 496	SLO 3

Student Outcome V

Outcome V: Developing an individual who possess a good familiarity with experimental and analytical methods, and techniques.

Metrics: The following metrics were used as targets of performance for outcome V.

- 1- The instructors' assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
- 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
- 3- 80% of Alumni surveyed expressed satisfaction with this outcome.
- 4- 70% of "co-op" and "Summer Internship" supervisors express favorable feedback on this outcome.
- 5- 70% satisfaction of "co-op" and "Summer Internship" students with mastery of this outcome.
- 6- 10% of CHE graduating seniors have participated in at least one "co-op" or "summer internship".

Supporting Courses and Activities:

Courses/activities	Elements Contributing to Outcome
CHE 239	SLOs 1-10
CHE 311	SLOs 2, 7, 8
CHE 317	SLOs 1, 4-6
CHE 418	SLOs 2-8
UG Research	Project research
COOP	Co-op Experience

Student Outcome VII

Outcome VII: Developing individuals who possess the ability to work in intra-disciplinary and multi-disciplinary teams and possess leadership skills.

Metrics: The following metrics were used as targets of performance for outcome VII.

- 1- The instructors' assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
- 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
- 3- 75% of the students respond favorably to question 4 on the exit interview form.
- 4- 80% of alumni surveyed expressed satisfaction with this outcome.
- 5- 70% of Co-op supervisors express favorable feedback on this outcome.
- 6- 70% satisfaction of co-op students with mastery of this outcome.
- 7- Favorable assessment by AIChE advisor.
- 8- College-wide IPRO assessment indicates satisfactory performance for this outcome.

Supporting Courses and Activities:

Course/Activity	Elements Contributing to Outcome
CHE 100	All SLOs
CHE 101	SLO 5
CHE 317	SLOs 1-10
IPRO 497	IPRO experience
CHE 418	SLOs 2-9
CHE 496	SLO 4
COOP	Co-op experience
AIChE	Chapter activities

Student Outcome IX

Outcome IX: Developing individuals who have the ability and desire to further their education through the process of “Life Long Learning” and possess the necessary skills to be involved in research and development activities, if they so desire.

- Metrics:** The following metrics were used as targets of performance for outcome IX.
- 1- The instructor’s assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
 - 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
 - 3- 75% of the students respond favorably to question 7 on the exit interview form.
 - 4- AIChE advisor reports satisfactory achievement of this outcome through student chapter activities and/or student involvement with national organization activities.
 - 5- College-wide IPRO assessment indicates satisfactory performance for this outcome.
 - 6- 80% of the alumni surveyed express satisfaction with this outcome.

Supporting Courses and Activities:

Course/Activity	Elements Contributing to Outcome
CHE 100	SLO 7 all?
CHE 101	SLOs 3
CHE 202	
CHE 301	
CHE 302	
CHE 311	SLO 2, 12
CHE 317	
CHE 351	
CHE 406	
CHE 423	
CHE 433	
CHE 435	
CHE 451	
CHE 494	
CHE 496	SLO 1
IPRO 497	Ethics Component
CHE 418	SLOs 1–8
AIChE	Chapter activities

Results

Evaluation of Student Outcomes

Evaluation of Student Outcome IV

Permanent Information

Outcome IV: Developing individuals who have been trained to fully utilize recent advances in computational technologies in their work.

Metrics: The following were used as targets of performance for outcome IV.

- 1- The instructors' assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
- 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
- 3- 75% of the students respond favorably to question 5 on the exit interview form.
- 4- 80% of alumni surveyed expressed satisfaction with this outcome.

Supporting Courses and Activities:

Courses/Activity	Elements Contributing to Outcome
CHE 302	SLO 6
CHE 494	SLOs 3 and 7

Outcome IV Evaluation

Metric 1:

All SLOs for CHE 302 and CHE 494 were met at > 70% level.

Of the courses examined this cycle there were no instructors indicating dissatisfaction with student knowledge in basic sciences or engineering. However, the course instructor for CHE 302 indicated deficiencies in mathematics. Specifically, many students appeared to show deficiencies in integration and solution of ordinary and partial differential equations.

Summary: Instructors indicated satisfaction with student knowledge in math, basic sciences for the academic year 2024- 2025, except for CHE 302, where deficiencies in mathematics were noted.

Metric 2:

This metric was met as > 70% of students indicated a mastery level for each SLO at 3 or above.

Metric 3:

Senior exit interviews are not being evaluated in this cycle.

Metric 3:

Alumni surveys are not being evaluated in this cycle

Observations for Outcome IV

Outcome IV: Developing individuals who have been trained to fully utilize recent advances in computational technologies in their work.

1. During the Fall 2024 semester, the course instructor for CHE 302 indicated deficiencies in mathematics,. In particular with integration and solving differential equations.
2. Exit interviews not evaluate this cycle.
3. Alumni surveys not evaluate this cycle

ABET Committee Recommendations for Outcome IV

1. Consider monitoring the proficiency of students in mathematics in future offerings of CHE 302 and other courses that rely on student's mastery of mathematics, in particular integration and solution of differential equations.

Evaluation of Student Outcome V

Permanent Information

Outcome V: Developing an individual who possess a good familiarity with experimental and analytical methods, and techniques.

Metrics: The following metrics were used as targets of performance for outcome V.

- 1- The instructors' assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
- 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
- 3- 80% of Alumni surveyed expressed satisfaction with this outcome.
- 4- 70% of "co-op" and "Summer Internship" supervisors express favorable feedback on this outcome.
- 5- 70% satisfaction of "co-op" and "Summer Internship" students with mastery of this outcome.
- 6- 10% of CHE graduating seniors have participated in at least one "co-op" or "summer internship".

Supporting Courses and Activities:

Course/Activity	Elements Contributing to Outcome
CHE 311	SLOs 2, 7, 8
CHE 317	SLOs 1, 4-6
CHE 418	SLOs 2-8

Outcome V Evaluation

Metrics 1 and 2:

Course instructor and student assessments for CHE 311, CHE 317 and CHE 418 indicate that >70% students were at adequate or above adequate performance for SLOs related to outcome V. One exception was for CHE 418 where SLO # 6 was not met because the crystallization experiment was out of commission during the Spring 2025 semester.

Summary: In general all Metrics1 and 2 were met at > 70% with one exception being SLO #6 in CHE 418 due to equipment not being available.

Metric 3:

Course instructors expressed satisfaction with the general knowledge in math, sciences and engineering principles of the students.

Summary: For Metric 3, all instructors were satisfied with student knowledge of the fundamentals of ChE

Metrics 4 and 5:

Senior exit interviews and alumni surveys are not being evaluated this cycle.

Observations for Outcome V

Outcome II: Developing an individual who possess a good familiarity with experimental and analytical methods, and techniques.

Evaluation of SLOs indicated areas where instructors need to focus more. No other concerns at this point in time except that maybe SLOs should be decoupled from laboratory experiments (e.g. SLO #6 in CHE 418).

ABET Committee Recommendations for Outcome V

1. Update course SLOs where needed or address a better way to evaluate the current SLOs. Specifically, SLOs for CHE 418 should not depend on availability of specific experiments (e.g. crystallization experiment in CHE 418).

Evaluation of Student Outcome VII

Permanent Information

Outcome VII: Developing individuals who possess the ability to work in intra-disciplinary and multi-disciplinary teams and possess leadership skills.

Metrics: The following metrics were used as targets of performance for outcome VII.

- 1- The instructors' assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
- 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
- 3- 75% of the students respond favorably to question 4 on the exit interview form.
- 4- 80% of alumni surveyed expressed satisfaction with this outcome.
- 5- 70% of Co-op supervisors express favorable feedback on this outcome.
- 6- 70% satisfaction of co-op students with mastery of this outcome.
- 7- Favorable assessment by AIChE advisor.
- 8- College-wide IPRO assessment indicates satisfactory performance for this outcome.

Supporting Courses and Activities:

Course/Activity	Elements Contributing to Outcome
CHE 317	SLOs 1-10
CHE 418	SLOs 2-9

Outcome VII Evaluation

Metrics 1 and 2:

Metrics 1 and 2 were met during the 2024-2025 cycle for the courses being evaluated (CHE-317, CHE-418).

Summary: All courses evaluated this cycle met metrics 1 and 2.

Metrics 4 and 5:

Senior exit interviews and alumni surveys are not being evaluated this cycle.

Observations for Outcome VII

No causes for concerns for Outcome VII.

ABET Committee Recommendations on Outcome VII

No action required.

Evaluation of Student Outcome IX

Permanent Information

Outcome IX: Developing individuals who have the ability and desire to further their education through the process of “Life Long Learning” and possess the necessary skills to be involved in research and development activities, if they so desire.

Metrics: The following metrics were used as targets of performance for outcome IX.

- 1- The instructor’s assessment for each SLO from each support course indicates that 70% of the students are at or above adequate performance.
- 2- 70% of the students enrolled in each support course rate their mastery level of each SLO supporting the outcome at 3 or above on the Student Self-Assessment SLO sheet.
- 3- 75% of the students respond favorably to question 7 on the exit interview form.
- 4- AIChE advisor reports satisfactory achievement of this outcome through student chapter activities and/or student involvement with national organization activities.
- 5- College-wide IPRO assessment indicates satisfactory performance for this outcome.
- 6- 80% of the alumni surveyed express satisfaction with this outcome.

Supporting Courses and Activities:

Course/Activity	Elements Contributing to Outcome
CHE 202	
CHE 302	
CHE 311	SLO 2, 12
CHE 317	
CHE 351	
CHE 451	
CHE 494	
CHE 418	SLOs 1–8

Outcome IX Evaluation

Metrics 1 and 2:

In the 2024-2025 academic year, more than 70% of the students enrolled in the courses listed in the previous table showed mastery of the SLOs at level 3 and above as indicated by student and instructor assessments. One exception was SLO # 6 in CHE 418 (Spring 2025 semester). This SLO was not evaluated because of a crystallization equipment that was out of commission during the Spring '25 semester.

Metrics 3 and 4:

Senior exit interviews and alumni surveys are not being evaluated this cycle.

Observations for Outcome IX

Outcome IX: Developing individuals who have the ability and desire to further their education through the process of “Life Long Learning” and possess the necessary skills to be involved in research and development activities, if they so desire

1. Only the evaluation of SLO 6 in CHE 418 could not be completed because of equipment that was not available during the Spring 2025 semester. In all courses > 70% students showed mastery of SLOs at levels 3 and above.

ABET Committee Recommendations on Outcome IX

1. All courses showed satisfactory accomplishment of SLO goals but it is recommended that SLOs do not become dependent on the availability of a single piece of equipment.

Artifact samples

Artifact samples are kept in file within the ChBE department and they can be made available upon request. Artifact examples are included in this report as a limited subset of the artifacts collected for all courses in the ChBE department. In all courses artifacts for the highest score, lower score and an average score are collected and archived in files for each assessment category (homeworks, quizzes, midterms, finals, student reports, presentations, etc). These can be accessed through request at the ChBE department.

Data table

The student outcomes of the ChBE department (**I – IX**) are evaluated across different courses. In each individual course, each student outcome may be supported by one or more Student Learning Objectives (SLOs) within each course. This makes the evaluation for each student outcome for each individual student cumbersome and it cannot be done within the current data collection process. This issue will be addressed in future reports.

Description of findings

In general, during the AY 2024 – 2025 the courses selected met the required metrics with the exception of CHE 418 where SLO # 6 was not met because of equipment being out of commission. The later is not major cause of concern but it indicates an area where course SLOs need to be modified. The examination of the data available yielded some insights on the useful of program evaluations. For some courses some metrics were not met earlier in the course as determined by quizzes but this allowed course instructors to address this deficiency such that the metric was met during the final exam. At the present time, and based on evaluation of courses during AY 2023 -124 there are no areas of concern other than the one noted for CHE 418. Some faculty have not returned the ABET evaluations, especially for Spring 2025 courses. The faculty should be more prompt in providing these evaluations in a timely manner.

Continuous Improvement, CI, plans

Currently there are no major areas of concern. A minor issue that became apparent with CHE 418 is that SLOs for lab courses should not be completely dependent on the operation status of a single piece of equipment. The UGSC and faculty teaching CHE 418 should consider revising the SLOs of courses where the availability of an equipment can determine whether a specific SLO is met or not. Student Learning Objectives (SLOs) should be periodically evaluated and modified if needed in order to keep the curriculum relevant. Data collection should be analyzed in order to determine if there are better ways to provide a comprehensive assessment of the curriculum.

Prior CI plan impact

There are no previous CI plans available. But in the self study report for ABET (2020), there were a set of recommendations made to continue improving the BS in Chemical Engineering program. Future assessments will yield more light on potential curricular modifications.

This cycle's CI plan

After consideration and presentation to the program authority, no CI plans are recommended.